docker compose unit 3

Convert the given file into docker compose :

```
FROM wordpress:latest
# Set environment variables for database connection
ENV WORDPRESS_DB_HOST=db
ENV WORDPRESS_DB_USER=wpuser
ENV WORDPRESS_DB_PASSWORD=wppassword
ENV WORDPRESS_DB_NAME=wpdatabase
# Expose port
EXPOSE 80
```

With docker run command :m

```
docker run -d \
--name wordpress_app \
-p 8082:80 \
-e WORDPRESS_DB_HOST=db \
-e WORDPRESS_DB_USER=wpuser \
-e WORDPRESS_DB_PASSWORD=wppassword \
-e WORDPRESS_DB_NAME=wpdatabase \
-v wordpress_data:/var/www/html \
mywordpress
```

```
PS C:\Users\amita> mkdir wordpressCompose


    Directory: C:\Users\amita


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
d-----        11-03-2026     12:20                wordpressCompose


PS C:\Users\amita> cd .\wordpressCompose\
PS C:\Users\amita\wordpressCompose> code .
PS C:\Users\amita\wordpressCompose>
```

File Edit Selection View Go ···

docker-compose.yml - wordpressCompose - Visual Studio Code

EXPLORER

OPEN EDITORS

Welcome

docker-compose.yml

WORDPRESSCOMPOSE

docker-compose.yml

OUTLINE

Welcome | docker-compose.yml

docker-compose.yml

Run All Services

```yaml
services:
  Run Service
  mywordpress:
    image: wordpress:latest
    container_name: wordpress_app
    environment:
      WORDPRESS_DB_HOST: db
      WORDPRESS_DB_USER: wpuser
      WORDPRESS_DB_PASSWORD: wppassword
      WORDPRESS_DB_NAME: wpdatabase
    ports:
      - "8082:80"
    volumes:
      - wordpress_data:/var/www/html

volumes:
  wordpress_data:
```

Images
Volumes
Kubernetes
Builds
Docker Hub
Docker Scout
Models
MCP Toolkit BETA
Extensions
Upgrade plan

Container CPU usage
1.13% / 1200% (12 CPUs available)

Container memory usage
149.8MB / 3.42GB

Show charts

Search

Only show running containers

Name	Container ID	Image	Port(s)	CPU (%)	Last s	Actions
interveiw	-	-	-	1.12%	27 mi	
wordpresscomp	-	-	-	0.01%	2 min	

Terminal

```
PS C:\Users\amita> cd wordpressCompose
PS C:\Users\amita\wordpressCompose> docker compose up -d
validating C:\Users\amita\wordpressCompose\docker-compose.yml: services.mywordpress.ports must be a array
PS C:\Users\amita\wordpressCompose> docker compose up -d
[+] up 28/28
 ✓ Image wordpress:latest                Pulled                                  39.0s
 ✓ Network wordpresscompose_default      Created                                  0.2s
 ✓ Volume wordpresscompose_wordpress_data Created                                 0.0s
 ✓ Container wordpress_app               Created                                  0.5s
PS C:\Users\amita\wordpressCompose>
```

Windo...

```
validating C:\Users\amita\wordpressCompose\docker-compose.yml: services.mywordpress.ports must be a array
PS C:\Users\amita\wordpressCompose> docker compose up -d
[+] up 28/28
 ✓ Image wordpress:latest                Pulled                                  39.0s
 ✓ Network wordpresscompose_default      Created                                  0.2s
 ✓ Volume wordpresscompose_wordpress_data Created                                 0.0s
 ✓ Container wordpress_app               Created                                  0.5s
PS C:\Users\amita\wordpressCompose> docker compose stop
[+] stop 1/1
 ✓ Container wordpress_app Stopped                                                1.6s
```

Practical No 4..>>>>>>>

Create the following folder structure:

```
node-mongo-compose/
│
├── docker-compose.yml
├── package.json
├── server.js
└── Dockerfile
```

file name : server.js

```
const express = require("express");
const mongoose = require("mongoose");
const app = express();
const PORT = 3000;
mongoose.connect("mongodb://mongo:27017/mydatabase", {
useNewUrlParser: true,
useUnifiedTopology: true
})
.then(() => console.log("MongoDB Connected"))
.catch(err => console.log(err));
app.get("/", (req, res) => {
res.send("Node.js + MongoDB running with Docker Compose");
});
app.listen(PORT, () => {
console.log(`Server running on port ${PORT}`);
});
```

File name : package.json

```
{
"name": "node-mongo-app",
"version": "1.0.0",
"main": "server.js",
"dependencies": {
"express": "^4.18.2",
"mongoose": "^7.0.0"
}
}
```

Dockerfile (for Node.js App)

```
FROM node:18
WORKDIR /app
COPY package.json .
RUN npm install
COPY . .
EXPOSE 3000
CMD ["node", "server.js"]
```

Docker Compose File

docker-compose.yml

```
version: '3.8'
services:
app:
build: .
container_name: node_app
ports:
- "3000:3000"
depends_on:
- mongo

restart: always
mongo:
image: mongo:6
container_name: mongo_db
ports:
- "27017:27017"
volumes:
- mongo_data:/data/db
restart: always
volumes:
mongo_data:
```

Steps to Run the Application

Step : Start containers

docker compose up –build

Step : Verify running containers

docker ps

docker-compose.yml

```yaml
version: '3.8'

Run All Services
services:
  Run Service
  app:
    build: .
    container_name: node_app
    ports:
      - "3000:3000"
    depends_on:
      - mongo
    restart: always

  Run Service
  mongo:
    image: mongo:6
    container_name: mongo_db
    ports:
      - "27017:27017"
    volumes:
      - mongo_data:/data/db
    restart: always

volumes:
  mongo_data:
```

Dockerfile > ...

```dockerfile
FROM node:18
WORKDIR /app
COPY package.json .
RUN npm install
COPY . .
EXPOSE 3000
CMD ["node", "server.js"]
```

{} package.json > ...

```
{
"name": "node-mongo-app",
"version": "1.0.0",
"main": "server.js",
"dependencies": {
"express": "^4.18.2",
"mongoose": "^7.0.0"
}
}
```

JS server.js > ...

```
const express = require("express");
const mongoose = require("mongoose");
const app = express();
const PORT = 3000;
mongoose.connect("mongodb://mongo:27017/mydatabase", {
useNewUrlParser: true,
useUnifiedTopology: true
})
.then(() => console.log("MongoDB Connected"))
.catch(err => console.log(err));
app.get("/", (req, res) => {
res.send("Node.js + MongoDB running with Docker Compose");
});
app.listen(PORT, () => {
console.log(`Server running on port ${PORT}`);
});
```

```
PS C:\Users\amita\My Project\docker\node-mongo-compose> docker compose up --build
time="2026-03-12T12:47:37+05:30" level=warning msg="C:\\Users\\amita\\My Project\\docker\\node-mongo-compo
se\\docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid
 potential confusion"
#1 [internal] load local bake definitions
#1 reading from stdin 575B done
#1 DONE 0.0s

#2 [internal] load build definition from Dockerfile
#2 transferring dockerfile: 151B 0.0s done
#2 DONE 0.0s
```

RAM 3.26 GB CPU 0.25% Disk: 62.53 GB used (limit 1006.85 GB)

Node.js + MongoDB running with Docker Compose

Question 1: The following configuration runs as Apache HTTP server.
version: “3”
Services:
Web:
Image:httpd
Modify the file to:

1. Map port 8081 on host to port 80 in container
2. Mount a local folder ./website to /usr/local/apache2/htdocs
3. Set container name as apache_server

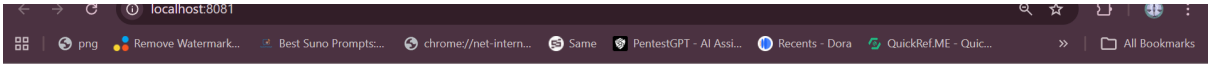
Question 2: Create a docker-compose.yml file to deploy the following application:
Requirements

1. Service 1: Nginx Web server

Image: nginx
Port Mapping: 8085:80

2. Service 2: Redis Database
   Image:redis
3. Both services must:
   Run on a custom network called appnetwork
4. The nginx container name should be nginx_server

```
tainer to be able to reuse that name.
PS C:\Users\amita\My Project\docker\ngnix-copose> ^C
PS C:\Users\amita\My Project\docker\ngnix-copose> docker rm -f apache_server
apache_server
PS C:\Users\amita\My Project\docker\ngnix-copose> docker-compose up -d
time="2026-03-13T12:45:10+05:30" level=warning msg="C:\\Users\\amita\\My Project\\docker\\ngnix-copose\\do
cker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid poten
tial confusion"
[+] up 1/1
 ✓ Container apache_server Started                                                    0.9s
PS C:\Users\amita\My Project\docker\ngnix-copose>
```



Index of /

docker-compose.yml ✕ Release Notes: 1.111.0

C: > Users > amita > My Project > docker > ngnix-copose > docker-compose.yml

```yaml
version: "3"
services:
  nginx:
    image: nginx
    container_name: nginx_server
    ports:
      - "8085:80"
    networks:
      - appnetwork

  redis:
    image: redis
    networks:
      - appnetwork

networks:
  appnetwork:
    driver: bridge
```

```
230e39d8034f   none                         null      local
a52f72bceca3   wordpresscompose_default     bridge    local
PS C:\Users\amita\My Project\docker\ngnix-copose> docker-compose up -d
time="2026-03-13T12:49:06+05:30" level=warning msg="C:\\Users\\amita\\My Project\\docker\\ngnix-copose\\do
cker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid poten
tial confusion"
[+] up 8/9
 ✓ Image redis                    Pulled                                                    11.2s
 - Network ngnix-copose_appnetwork Creating                                                 0.1s
[+] up 11/113-13T12:49:17+05:30" level=warning msg="Found orphan containers ([apache_server]) for this pro
 ✓ Image redis                    Pulled                                                    11.2s
```

Windo...

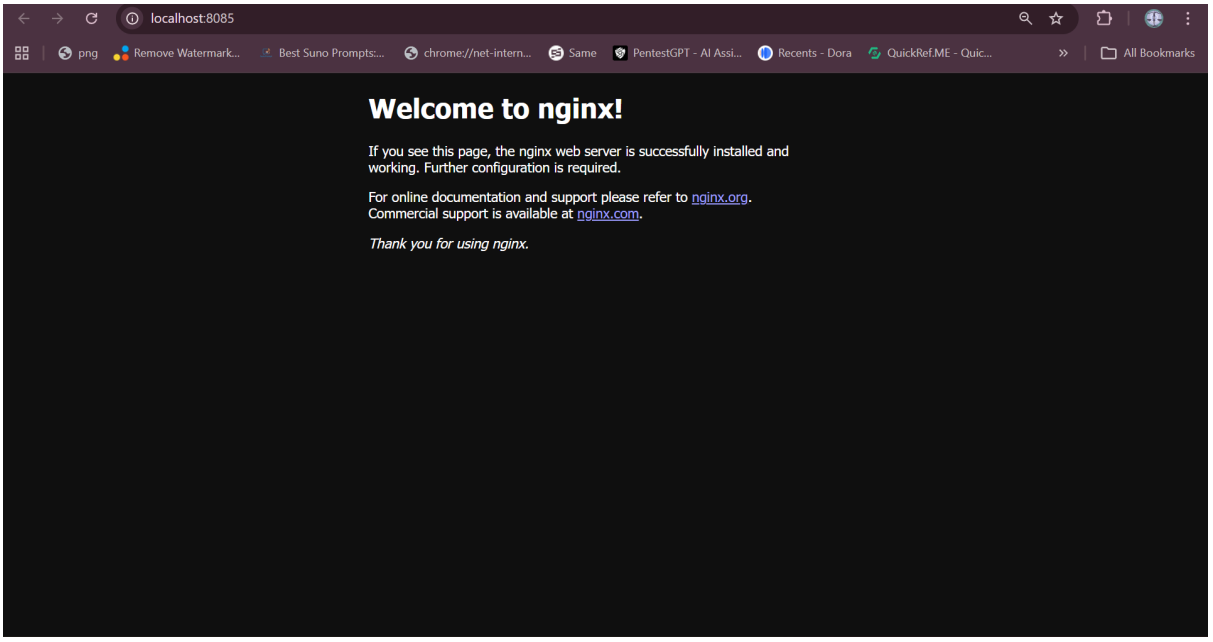
RAM 1.57 GB CPU 0.25% Disk: 62.59 GB used (limit 1006.85 GB)

```
>>
>> # Inspect which containers are on appnetwork
>> docker network inspect appnetwork
CONTAINER ID   IMAGE                    COMMAND                  CREATED              STATUS
PORTS                                                   NAMES
fd3a81c80507   nginx                    "/docker-entrypoint.…"   About a minute ago   Up About a minute
0.0.0.0:8085->80/tcp, [::]:8085->80/tcp          nginx_server
ca7123deda30   redis                    "docker-entrypoint.s…"   About a minute ago   Up About a minute
6379/tcp                                         ngnix-copose-redis-1
d84772bf02f0   httpd                    "httpd-foreground"       5 minutes ago        Up 5 minutes
0.0.0.0:8081->80/tcp, [::]:8081->80/tcp          apache_server
```

RAM 1.32 GB CPU 0.67% Disk: 62.59 GB used (limit 1006.85 GB)

```
PS C:\Users\amita\My Project\docker\ngnix-copose> docker ps
CONTAINER ID   IMAGE                    COMMAND                  CREATED             STATUS
PORTS                                                   NAMES
fd3a81c80507   nginx                    "/docker-entrypoint.…"   About a minute ago   Up About a minute
0.0.0.0:8085->80/tcp, [::]:8085->80/tcp                 nginx_server
ca7123deda30   redis                    "docker-entrypoint.s…"   About a minute ago   Up About a minute
6379/tcp                                                ngnix-copose-redis-1
d84772bf02f0   httpd                    "httpd-foreground"       6 minutes ago        Up 6 minutes
0.0.0.0:8081->80/tcp, [::]:8081->80/tcp                 apache_server
740cb784f755   node-mongo-compose-app   "docker-entrypoint.s…"   24 hours ago         Up 28 minutes
RAM 1.32 GB  CPU 1.25%   Disk: 62.59 GB used (limit 1006.85 GB)
```

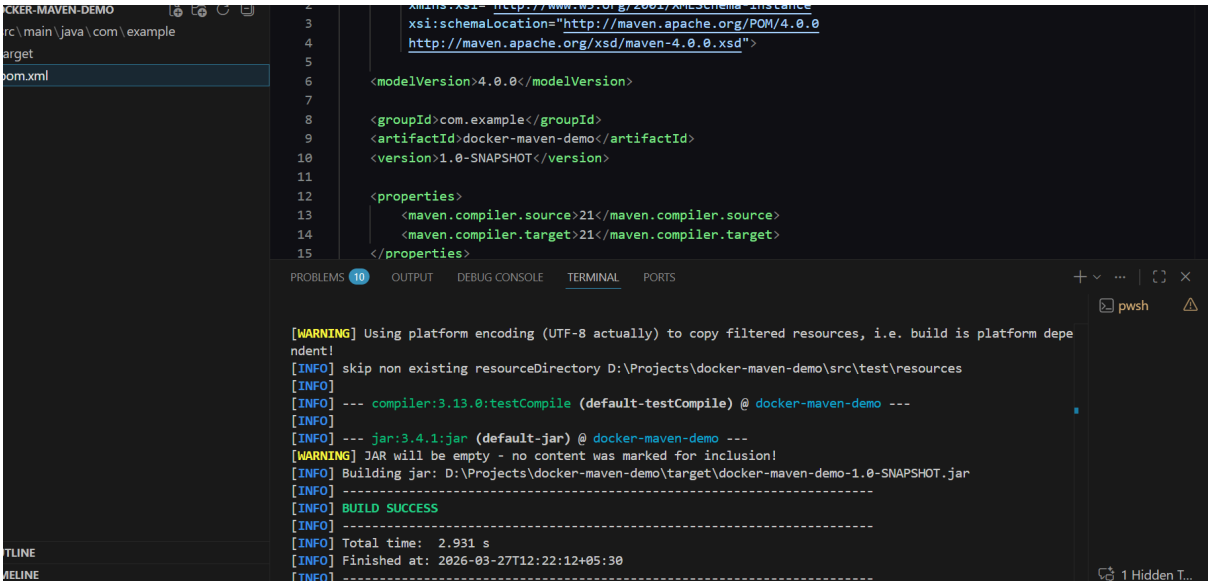
```
PS C:\Users\amita\My Project\docker\ngnix-copose> docker network ls
NETWORK ID     NAME                      DRIVER    SCOPE
693c18ab21ab   amiai_default             bridge    local
fadcd45eefa7   bridge                    bridge    local
e246c576fc12   docker-compose_default    bridge    local
02c724f7efcf   gabizap-network           bridge    local
c5bbd7727bdd   host                      host      local
49921a945773   interveiw_default         bridge    local
ecad051b27d9   ngnix-copose_appnetwork   bridge    local
26b11deeb9fb   ngnix-copose_default      bridge    local
a199cb9046e4   node-mongo-compose_default bridge   local
RAM 1.32 GB  CPU 0.25%   Disk: 62.59 GB used (limit 1006.85 GB)
```

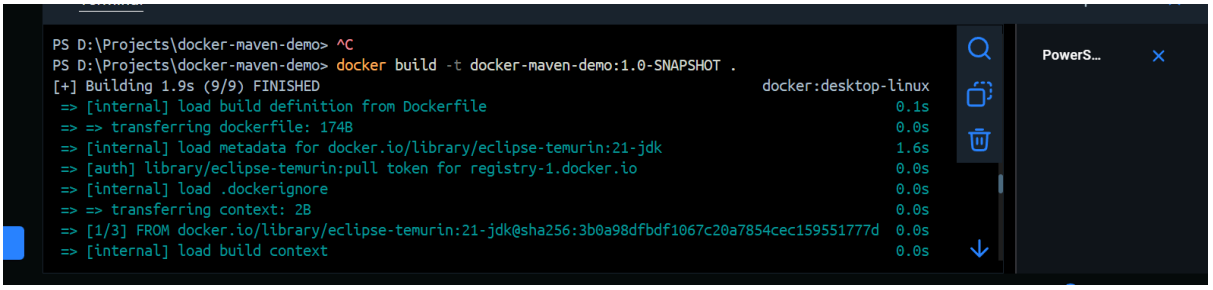


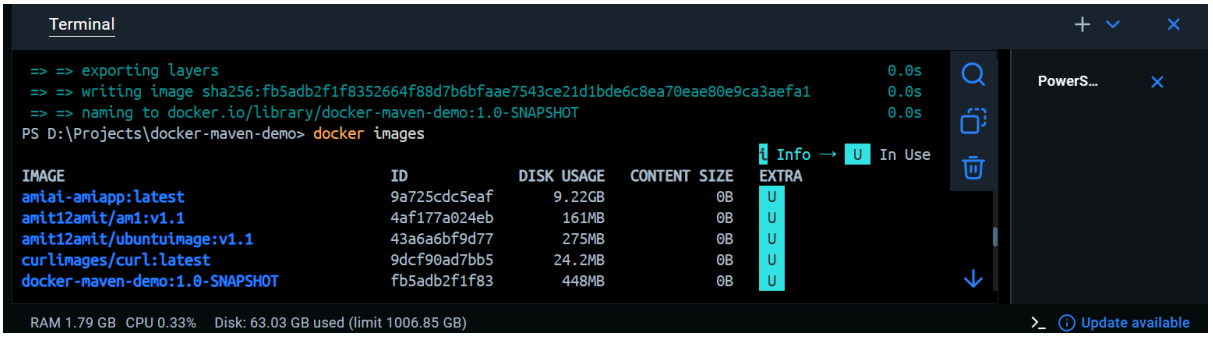
Tab 2

Unit 4

Docker + maven







Practical 3:

```
C:\Users\amita>mvn archetype:generate -DgroupId=com.demo -DartifactId=web-demo -DarchetypeArtifactId=maven-archetype-webapp -Dinterac
tiveMode=false
WARNING: A terminally deprecated method in sun.misc.Unsafe has been called
WARNING: sun.misc.Unsafe::staticFieldBase has been called by com.google.inject.internal.aop.HiddenClassDefiner (file:/D:/Main/apache-
maven-3.9.11-bin/apache-maven-3.9.11/lib/guice-5.1.0-classes.jar)
WARNING: Please consider reporting this to the maintainers of class com.google.inject.internal.aop.HiddenClassDefiner
WARNING: sun.misc.Unsafe::staticFieldBase will be removed in a future release
[INFO] Scanning for projects...
Downloading from central: https://repo.maven.apache.org/maven2/org/codehaus/mojo/maven-metadata.xml
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-metadata.xml
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-metadata.xml (14 kB at 18 kB/s)
Downloaded from central: https://repo.maven.apache.org/maven2/org/codehaus/mojo/maven-metadata.xml (21 kB at 25 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-archetype-plugin/maven-metadata.xml
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-archetype-plugin/maven-metadata.xml (1.1
 kB at 7.3 kB/s)
[INFO]
[INFO] ------------------< org.apache.maven:standalone-pom >-------------------
[INFO] Building Maven Stub Project (No POM) 1
[INFO] --------------------------------[ pom ]---------------------------------
[INFO]
[INFO] >>> archetype:3.4.1:generate (default-cli) > generate-sources @ standalone-pom >>>
[INFO]
[INFO] <<< archetype:3.4.1:generate (default-cli) < generate-sources @ standalone-pom <<<
[INFO]
[INFO]
[INFO] --- archetype:3.4.1:generate (default-cli) @ standalone-pom ---
[INFO] Generating project in Batch mode
```

```
[INFO] Using following parameters for creating project from Old (1.x) Archetype: maven-archetype-webapp:1.0
[INFO] ----------------------------------------------------------------------------
[INFO] Parameter: basedir, Value: C:\Users\amita
[INFO] Parameter: package, Value: com.demo
[INFO] Parameter: groupId, Value: com.demo
[INFO] Parameter: artifactId, Value: web-demo
[INFO] Parameter: packageName, Value: com.demo
[INFO] Parameter: version, Value: 1.0-SNAPSHOT
[INFO] project created from Old (1.x) Archetype in dir: C:\Users\amita\web-demo
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  10.424 s
[INFO] Finished at: 2026-03-31T12:27:32+05:30
[INFO] ------------------------------------------------------------------------

C:\Users\amita>cd web-demo

C:\Users\amita\web-demo>web-demo/src/main/webapp/index.jsp
'web-demo' is not recognized as an internal or external command,
operable program or batch file.
```

```
C:\Users\amita\web-demo>mvn clean install
WARNING: A terminally deprecated method in sun.misc.Unsafe has been called
WARNING: sun.misc.Unsafe::staticFieldBase has been called by com.google.inject.internal.aop.HiddenClassDefiner (file:/D:/Main/apache-
maven-3.9.11-bin/apache-maven-3.9.11/lib/guice-5.1.0-classes.jar)
WARNING: Please consider reporting this to the maintainers of class com.google.inject.internal.aop.HiddenClassDefiner
WARNING: sun.misc.Unsafe::staticFieldBase will be removed in a future release
[INFO] Scanning for projects...
[INFO]
[INFO] -------------------------< com.demo:web-demo >--------------------------
[INFO] Building web-demo Maven Webapp 1.0-SNAPSHOT
[INFO]   from pom.xml
[INFO] --------------------------------[ war ]---------------------------------
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-war-plugin/3.4.0/maven-war-plugin-3.4.0
.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-war-plugin/3.4.0/maven-war-plugin-3.4.0.
pom (8.4 kB at 10 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-war-plugin/3.4.0/maven-war-plugin-3.4.0
.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-war-plugin/3.4.0/maven-war-plugin-3.4.0.
jar (83 kB at 492 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/junit/junit/3.8.1/junit-3.8.1.pom
Downloaded from central: https://repo.maven.apache.org/maven2/junit/junit/3.8.1/junit-3.8.1.pom (998 B at 13 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/junit/junit/3.8.1/junit-3.8.1.jar
Downloaded from central: https://repo.maven.apache.org/maven2/junit/junit/3.8.1/junit-3.8.1.jar (121 kB at 1.2 MB/s)
[INFO]
[INFO] --- clean:3.2.0:clean (default-clean) @ web-demo ---
[INFO]
[INFO] --- resources:3.3.1:resources (default-resources) @ web-demo ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] Copying 0 resource from src\main\resources to target\classes
[INFO]
[INFO] --- compiler:3.13.0:compile (default-compile) @ web-demo ---
[INFO] No sources to compile
```

```
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  13.198 s
[INFO] Finished at: 2026-03-31T12:32:29+05:30
[INFO] ------------------------------------------------------------------------

C:\Users\amita\web-demo>mvn tomcat7:run
WARNING: A terminally deprecated method in sun.misc.Unsafe has been called
WARNING: sun.misc.Unsafe::staticFieldBase has been called by com.google.inject.internal.aop.HiddenClassDefiner (file:/D:/Main/apache-
maven-3.9.11-bin/apache-maven-3.9.11/lib/guice-5.1.0-classes.jar)
WARNING: Please consider reporting this to the maintainers of class com.google.inject.internal.aop.HiddenClassDefiner
WARNING: sun.misc.Unsafe::staticFieldBase will be removed in a future release
[INFO] Scanning for projects...
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/tomcat/maven/tomcat7-maven-plugin/2.2/tomcat7-maven-plugin-
2.2.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/tomcat/maven/tomcat7-maven-plugin/2.2/tomcat7-maven-plugin-2
.2.pom (12 kB at 25 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/tomcat/maven/tomcat-maven-plugin/2.2/tomcat-maven-plugin-2.
2.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/tomcat/maven/tomcat-maven-plugin/2.2/tomcat-maven-plugin-2.2
.pom (33 kB at 114 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/apache/12/apache-12.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/apache/12/apache-12.pom (16 kB at 80 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/tomcat/maven/tomcat7-maven-plugin/2.2/tomcat7-maven-plugin-
2.2.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/tomcat/maven/tomcat7-maven-plugin/2.2/tomcat7-maven-plugin-2
.2.jar (114 kB at 575 kB/s)
[INFO]
[INFO] ------------------------< com.demo:web-demo >-------------------------
[INFO] Building web-demo Maven Webapp 1.0-SNAPSHOT
```

```
Downloading from central: https://repo.maven.apache.org/maven2/org/slf4j/jcl-over-slf4j/1.7.5/jcl-over-slf4j-1.7.5.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-interactivity-api/1.0-alpha-4/plexus-interac
tivity-api-1.0-alpha-4.jar (13 kB at 4.4 kB/s)
Downloaded from central: https://repo.maven.apache.org/maven2/org/sonatype/plexus/plexus-build-api/0.0.4/plexus-build-api-0.0.4.jar (
6.8 kB at 2.2 kB/s)
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/maven-plugin-descriptor/2.2.1/maven-plugin-descriptor-
2.2.1.jar (39 kB at 12 kB/s)
Downloaded from central: https://repo.maven.apache.org/maven2/org/slf4j/jcl-over-slf4j/1.7.5/jcl-over-slf4j-1.7.5.jar (17 kB at 5.0 k
B/s)
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/maven-settings/2.2.1/maven-settings-2.2.1.jar (49 kB a
t 14 kB/s)
[INFO] Running war on http://localhost:8080/
[INFO] Creating Tomcat server configuration at C:\Users\amita\web-demo\target\tomcat
[INFO] create webapp with contextPath:
Mar 31, 2026 12:32:58 PM org.apache.coyote.AbstractProtocol init
INFO: Initializing ProtocolHandler ["http-bio-8080"]
Mar 31, 2026 12:32:58 PM org.apache.catalina.core.StandardService startInternal
INFO: Starting service Tomcat
Mar 31, 2026 12:32:58 PM org.apache.catalina.core.StandardEngine startInternal
INFO: Starting Servlet Engine: Apache Tomcat/7.0.47
Mar 31, 2026 12:32:59 PM org.apache.coyote.AbstractProtocol start
INFO: Starting ProtocolHandler ["http-bio-8080"]
```

practicAL 4;